

ArtSpeak: An Interactive AR Application for Lifelike Speaking with Art Portraits

Shubhangi S. R. Garnaik^{1,2*}

Aruna Balasubramanian^{2†}

Niranjan Balasubramanian^{2‡}

Jihoon Ryoo^{1,2§}

¹Department of Computer Science, The State University of New York, Korea
²Department of Computer Science, Stony Brook University, USA

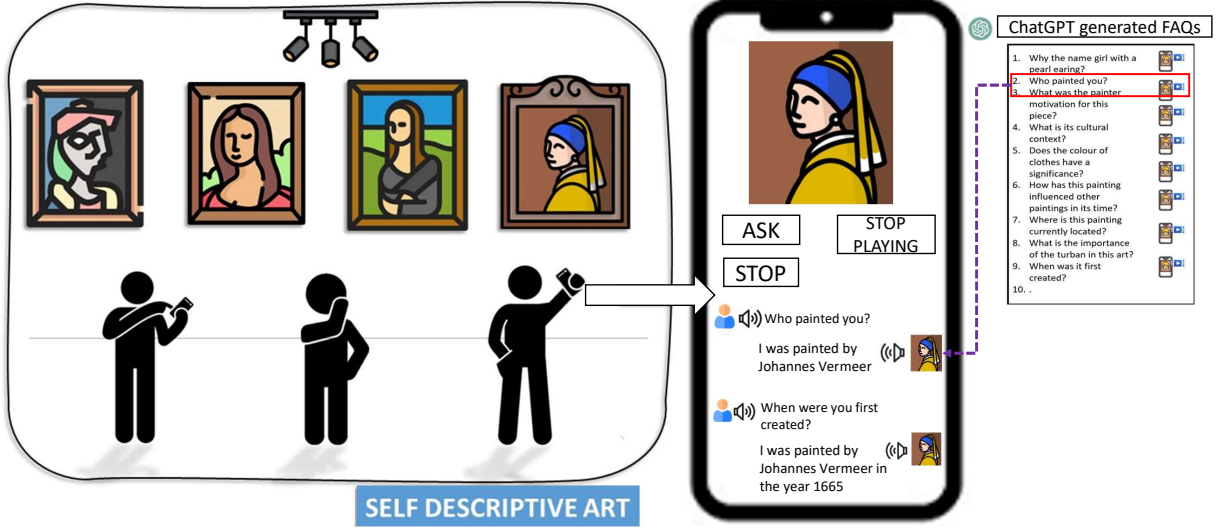


Figure 1: System overview of *ArtSpeak*, a mobile AR application that enables museum visitors to interact with art portraits through user queries. Visitors point their phone at a portrait and ask questions, receiving lifelike talking-head video responses. The system retrieves answers from a curated set of frequently asked questions (FAQs) generated using ChatGPT, to enable personalized interactions.

ABSTRACT

Museum visits often lack personalized and interactive experiences, limiting visitor engagement with art and historical artifacts. To address this, we present *ArtSpeak*, a standalone augmented reality (AR) application that transforms traditional art viewing into an interactive storytelling experience. When users point their mobile cameras at an artwork, the system responds to their questions with lifelike, talking-head video narratives generated from historical portraits. However, generating such talking-head videos at runtime is computationally expensive, often requiring over a minute per response. To address this challenge, *ArtSpeak* introduces two major contributions. First, it employs a collection of frequently asked questions (FAQ) to generate a set of lifelike video responses for various art portraits. Second, it introduces a novel retrieval-based approach that uses GPT-based embeddings and cosine similarity to select the most relevant response.

*e-mail: shubhangisaile.garnaik@stonybrook.edu

ORCID: 0009-0005-3211-1703

†e-mail: arunab@cs.stonybrook.edu

ORCID: 0000-0003-3720-2215

‡e-mail: niranjan@cs.stonybrook.edu

ORCID: 0000-0003-4187-9368

§e-mail: jihoon.ryoo@sunykorea.ac.kr

ORCID: 0000-0002-8330-8347

As a result, the system dynamically presents the video reply that best aligns with the user’s inquiry, reducing computational overhead and ensuring a real-time, low-latency experience. More precisely, *ArtSpeak* achieves over $30\times$ lower latency and reduces energy consumption by approximately 81% compared to the real-time video generation method. User studies further validate the system’s effectiveness, with 85% of participants rating the retrieved responses as relevant to their queries and 90% reporting smooth video playback. These results highlight the efficiency and user satisfaction enabled by our retrieval-based approach.

Index Terms: Augmented Reality, Generative AI, Cultural Heritage Interfaces, FAQ Retrieval, Human-Centered AI, Natural Language Interaction, Mobile AR Application.

1 INTRODUCTION

Since the 19th century, public access to private art collections helped establish museum visits as a popular leisure activity. However, shifting lifestyles, reduced leisure time, and external factors such as the COVID-19 pandemic have challenged traditional museum experiences, leading to a decline in visitor numbers. According to a survey conducted by the Art Newspaper, there was a staggering 77% decrease in visits to the 100 most popular art museums worldwide in 2020[9, 5], thus emphasizing the need for creative solutions to restore museum attendance and engagement. Modern visits are often limited by time-consuming tours, limited contextual information, and difficulty engaging with static displays, highlighting the need for more flexible and enriching visitor experiences. To address these issues, we propose a novel AR application, that in-

tegrates Deep Neural Networks (DNN), Augmented Reality (AR), and Natural Language Processing (NLP) to provide personalized museum experiences. Natural Language Processing (NLP) techniques such as information retrieval and question answering, integrated with DNN models[37, 27] have transformed applications across domains like media, healthcare[48, 56], and tourism, ultimately enriching human experiences.

Current research has explored AR and VR applications [15, 22] within museums and art galleries. These include marker-based, markerless, and positioning-based AR experiences, offering diverse interactions for users across age groups. However, existing systems often lack personalization due to static responses, do not support conversational interfaces, and overlook latency challenges critical for real-time deployment. Our prior work on adaptive video quality [3, 1, 2, 36] highlights the importance of context-awareness, perceptual optimization, and low-latency delivery—principles that can be effectively extended to immersive AR/VR systems in cultural settings. With further technological advancement and creativity, similar innovations as [21] could help reimagine the future of culture and education.

In this work, we design a mobile AR application that transforms passive art viewing into an interactive storytelling experience. When users point their mobile cameras at an art portrait, the portrait responds to the user’s question with a lifelike, talking-head video. As shown in Figure. 1, the application is designed to be easily accessible to museum visitors, without additional infrastructure. Users engage with the artwork by asking questions and receiving personalized responses. To enable this smooth interaction, the system relies on a curated set of frequently asked questions (FAQs) that models visitor inquiries, generated using ChatGPT.

There are several research challenges in the deployment of generative models in real-time AR systems. These models were computationally intensive and took more than a minute to generate realistic talking-head portrait videos. To address latency issues, we introduced our retrieval-based pipeline and pregenerated lifelike talking head videos of art portraits for curated FAQ responses—generated using ChatGPT from generative AI models. These videos are stored in a local database and dynamically retrieved at runtime. When the system receives the user’s query, the retrieval-based system semantically matches it to the relevant FAQs and generates a response by retrieving or combining the most appropriate video response segments. This approach enables low-latency, conversational interactions without an on-device inference at runtime.

We evaluated the *ArtSpeak* system through both a user study and quantitative analysis using latency and retrieval metrics. The user study involved 23 participants with varying levels of familiarity with AR technologies and different museum visitation habits (weekly, monthly or yearly). Participants interacted with the system using a range of user queries and rated their experience based on system responsiveness, response smoothness, and overall experience. In addition, we compared various FAQ retrieval approaches by evaluating different embedding models and indexing techniques. Our method *GPT-based embeddings with cosine similarity and dictionary indexing* achieved the best overall performance. The results confirm that *ArtSpeak* response times up to $30\times$ faster than real-time talking head video generation, with 85% of users reporting useful and relevant interactions.

Our system demonstrates how AI-generated content and NLP-driven information retrieval can revolutionize museum experiences with personalized and dynamic responses to encourage interest in cultural heritage. Details regarding the system flow and its components along with the user evaluation study are described next.

2 RELATED WORKS

Several research studies have explored ways in which AR technology could improve museum storytelling, accessibility, and engagement. This section examines important lines of related work, such as social interaction systems, augmented reality (AR) experiences, gamified interfaces, audio-based upgrades, and AI-driven virtual guides to identify constraints that our system addresses in this work.

Museums have been spaces for cultural preservation and public education. Traditionally, people used guides, brochures, and on-line resources to learn about artifacts. With the rise of mobile and social technologies, many researchers have explored accessibility and inclusivity, especially for visitors with disabilities. During the mid-2010s, the introduction of systems such as Instagram-powered museum engagements and Access+ [50, 40, 28, 46, 16, 18], which relied on social media to encourage interaction and accessibility. However, these systems lacked conversational range and real-time customization and mostly used text or static material.

The integration of augmented and virtual reality into museum applications has revolutionized the way cultural heritage is presented. AR enhances engagement by overlaying interactive digital content on to physical artifacts, offering immersive and educational experiences[26, 58, 33]. Systems such as Variable Museums and MoSA explored marker-based AR to display virtual artworks or highlight stolen art histories[8, 12]. Marker-less AR systems, including those used in the MeLa project [17], [47, 4, 45, 38, 25], use natural feature tracking to reveal hidden aspects or reconstruct historic contexts. The ARCO system [51] enabled 3D artifact models to create VR/AR shows, however it only provides visual augmentations and is dependent on in-museum infrastructure. Similarly, the MARSS project[41] used AR to present 3D reconstructions of ancient instruments with animated panels and touch-based interaction, improving engagement in science education but lacking mobile or dialog-based interaction.

A growing body of work investigates Audio Augmented Reality (AAR) for immersive, non-visual engagement. The Audible Artefact project used audio overlays and locative soundscapes to engage visitors with archived sounds and artifacts through ambient storytelling [10, 52, 14, 31]. Similarly, Mapping the Symphony and Listening Sessions at the National Science and Media Museum emphasized the interpretive power of sound to foreground auditory storytelling. These systems expand narrative modalities but do not address visual interaction or natural language queries.

Additionally, some research explores Audio Augmented Reality (AAR) for immersive, non-visual engagement in museums. The Audible Artefact project [10, 39, 23] used locative soundscapes and audio overlays for visitors to interact with archived sounds and cultural artifacts through ambient storytelling. Similarly, the National Science and Media Museum’s Mapping the Symphony and the Listening Sessions emphasize the curatorial experience’s ability to understand sound. Although these systems extended narrative modes, they lack natural language questions or visual interaction.

Furthermore, AI-driven virtual humans have emerged as a novel interaction modality. Spyrou [43, 6] proposed a reference architecture for the GLAM (Galleries, Libraries, Archives, Museums) sector that integrates AI-powered NPC guides into a virtual metaverse gallery. Built using Unreal Engine and Metahuman Creator, the system enables immersive storytelling through lifelike avatars. However, such approaches depend on infrastructure-intensive environments such as VR headsets, meta-verse platforms and are difficult to setup for changing museum settings. Similarly, a mixed-reality holographic guide was introduced inside a HoloLens [20]; however, it depends on costly headset fleets and per-persona volumetric capture. Spallone and team’s research [42] introduces AR-heritage avatars with a strong spatial presence on a handheld AR device as a museum guide, yet each statue still needs an expensive

Table 1: SADTALKER inference time across hardware configurations.

Portrait	Duration (mm:ss)	Laptop (i7)	Xeon E5	i9 + GPU
Laughing Cavalier	00:17	03:33	01:17	00:52
Monalisa	00:23	04:59	01:48	01:12
Adele	00:18	03:41	01:22	00:55
Blue Boy	00:20	04:30	01:40	01:06
Girl with Pearl	00:15	03:20	01:12	00:48
Frida – Thorn	00:14	03:06	01:07	00:45

3-D pipeline: photogrammetry, rigging, and key-framed animation, thereby raising authoring cost. Large-language-model (LLM) chat-bots have recently been paired with 2-D avatar guides [54], but they lack lip-sync and rely on bubble text or audio-only narration, which limits immersion. Furthermore, the VirtueWander [49] system offers multimodal guidance through a single guide avatar, limiting interaction and visitors’ ability to explore specific artworks in museums.

Despite remarkable advances across AR, VR, AAR, and virtual avatar systems, most existing solutions share key limitations: reliance on fixed or high-cost infrastructure, delivery of static content, visual quality, inference speed, limited cross-identity generalization, multiple pre-processing steps, system complexity and lack of real-time, dialog-driven personalization.

Our work addresses these gaps through a lightweight, mobile AR application that enables visitors to interact directly with physical artworks via expressive, animated portraits. By combining a generative talking-head model (SADTALKER) with an NLP-powered FAQ retrieval system, we create a dialog-based storytelling experience that is personalized, responsive, and requires no changes to the museum environment. This approach transforms passive art observation into an interactive, conversational engagement, offering a scalable and accessible solution for cultural spaces.

3 SYSTEM OVERVIEW

3.1 Challenges

Museum visitors usually expect responsive and immersive digital experiences. However, achieving this in real-time using animated realistic talking portraits presents two key challenges: First, generating lifelike, talking-head videos using models like SADTALKER is computationally expensive. Even on GPU-enabled desktops, each video can take more than a minute to generate, while on standard laptops, this time can exceed six minutes, as shown in Table 1. Hence, real-time video generation is impractical for mobile applications due to this latency, which could result in a decline in user engagement. Second, most existing information retrieval systems are built for specific domains and do not generalize well in the context of museum art portraits. When used in this setting, they often fail to understand the context of visitor questions, thus lead to irrelevant responses. This can reduce user engagement and limits the educational value of the interaction.

To address these limitations, we built *ArtSpeak*, an augmented reality (AR) mobile application that enables conversational interaction with museum artworks. The two main components of the system are, a FAQ retrieval approach that matches user questions to carefully chosen museum-specific FAQ answers based on semantics, and a pre-generation pipeline that uses the SADTALKER model to animate portraits with realistic video responses. This application is scalable for real-world museum deployment because it retrieves and combines videos at runtime instead of creating them in real-time, ensuring low-latency and contextually relevant interactions.

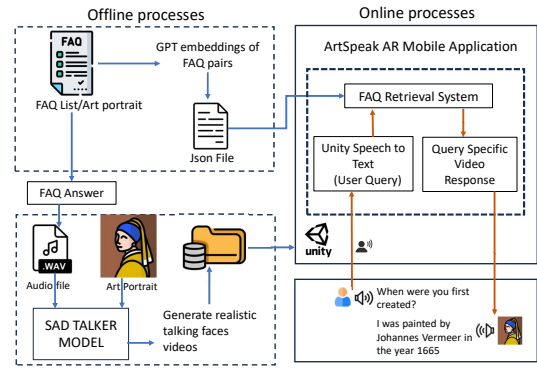


Figure 2: System architecture of *ArtSpeak*, consists of an offline phase for generating FAQ embeddings and pre-generated talking-head videos using the SADTALKER model, and an online phase where user queries are semantically matched to FAQs and corresponding videos are retrieved for AR-based playback.

3.2 System flow

This section explains the system architecture and operational flow of *ArtSpeak*. The system workflow consists of offline and on-line processes, as illustrated in Figure. 2

In the offline process, two main tasks are carried out:

1. **FAQ Generation and Embedding:** We generate a collection of frequently asked questions (FAQs) specific to each piece of art, to allow personalized interactions. These FAQs serve as the basis for retrieving semantically relevant responses to user queries while visiting museums. The FAQs cover common visitor questions about each art portrait, including details about the artist, their origin, attire, and historical significance. We used OpenAI’s ChatGPT to generate initial question-answer pairs, which were further revised by reviewing and filtering out repetitive or subset FAQs covered by generic FAQs. Sentence embedding models were applied to encode each pair of FAQs into embeddings, which were stored for efficient lookup in the retrieval system.
2. **Lifelike Portrait Video Generation:** The answers for each FAQ are converted into audio files using a text-to-audio model. These audio files and art portrait images are fed into the SADTALKER model to produce realistic talking-face videos. Real-time video generation takes over a minute even on high-end systems and up to 6 minutes on standard laptops, as shown in Table 1. To reduce latency and ensure smooth user experience, all videos are pre-generated using SADTALKER and stored in a database for on-demand playback within the AR application.

The online process involves the *ArtSpeak* application operating in real time. When the user asks a question, the system converts it to text using a speech-to-text module. This transcribed query is then passed to the FAQ retrieval module where it identifies the most relevant pre-generated video response and displays it on the user’s mobile device.

4 FAQ RETRIEVAL SYSTEM

To improve user interaction within the *Artspeak* application, we developed a robust FAQ retrieval system as shown in Figure. 3. It uses Large Language Models (LLMs) to generate sentence-level embeddings and enable real-time retrieval. The system was designed to answer user queries regarding specific art portraits, providing responses that were contextually related.

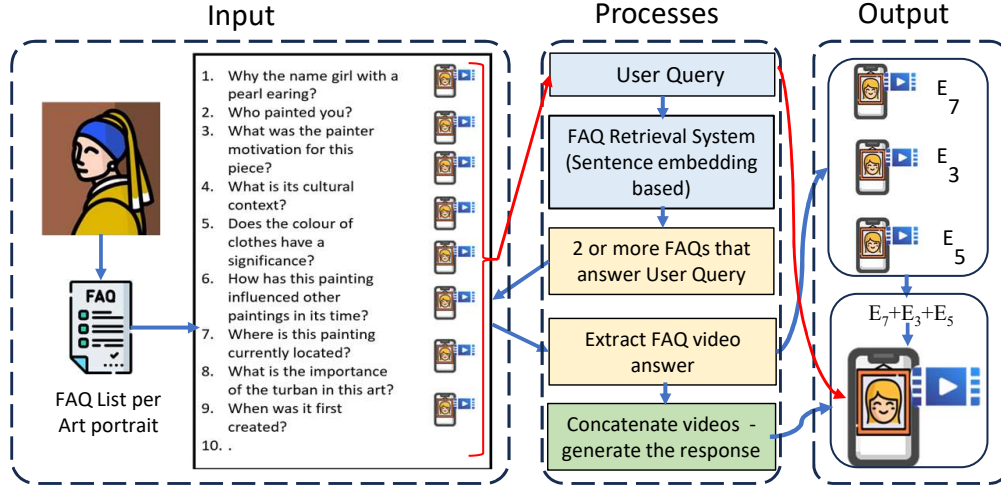


Figure 3: FAQ-based response generation pipeline. User queries are matched to relevant FAQs, and corresponding video segments are retrieved and merged to form the final response.

4.1 Generating FAQs by Persona and Aspect

In this study, we considered a total of six art portraits, for each of them, we generated nearly 140 FAQ using ChatGPT. We used a structured prompt-based approach to create these FAQs, across 20 comprehensive art interpretation aspects and replicating user perspectives through 15 different visitor personas. These personas were considered to represent a wide range of museum visitors, such as students, art historians, tourists, and children, as shown in Table 2. This would help the system prepare for a wide range of information requirements and query types.

All generated questions and answers were manually reviewed to remove duplicates, rephrased, and validated for contextual precision. This approach supports a scalable and flexible response system. In an actual museum deployment, new FAQs may be created and integrated using this organized process as the collection of supported art portraits grows. Moreover, as structured prompts and diverse visitor personas are considered to generate FAQs, the system can generalize across a wide range of artworks and effectively capture several user intents.

4.2 Embedding and Storage for Retrieval

These FAQs were embedded as sentence representations for effective information retrieval based on the semantics of user queries. For a given FAQ set (Q_i, A_i) , the embedding vector E_i is computed as:

$$E_i = f(Q_i \oplus A_i) \quad (1)$$

where:

- Q_i : The i -th question text.
- A_i : The i -th answer text.
- f : The embedding function specific to the model (e.g., SBERT, USE, GPT).
- $\oplus$: Concatenation operator.

We evaluated three widely used sentence embedding models: Sentence-BERT (SBERT) [34], Google Universal Sentence Encoder (USE) [7], and OpenAI’s GPT-based embeddings [30]. These models were chosen to balance semantic accuracy (SBERT, GPT), inference efficiency (USE), and deployability (SBERT locally, GPT

Table 2: List of visitor personas and interpretive aspects used to generate diversified FAQs for each artwork.

Personas	Aspects
1. Art Enthusiasts	1. Artist Information
2. Tourists	2. Title and Date
3. Families	3. Subject and Identity
4. Students	4. Composition
5. Casual Visitors	5. Style and Genre
6. Art Collectors	6. Techniques and Medium
7. Cultural Explorers	7. Size and Scale
8. Art Students and Aspiring Artists	8. Color Palette
9. Seniors	9. Background and Setting
10. Art Critics and Reviewers	10. Symbolism and Imagery
11. Event Attendees	11. Emotion and Expression
12. Researchers and Historians	12. Lighting and Shadow
13. Art Therapy Participants	13. Brushwork and Texture
14. Art Market Professionals	14. Historical and Cultural Context
15. Local Residents	15. Provenance and Ownership
	16. Critical Reception
	17. Restoration and Conservation
	18. Artistic Influences and References
	19. Viewer Interpretation
	20. Legacy and Impact

using API) and show realistic trade-offs for real-time applications. They are better at capturing contextual meanings more effectively compared to simple keyword matching and thus can improve the quality of the information retrieval system. FAQ embeddings of these models were stored in structured files, for efficient lookup and ranking during real-time user interaction.

At runtime, the user’s spoken input is transcribed to text and converted into an embedding vector $\mathbf{u}$. The system computes cosine similarity between this vector and each precomputed FAQ embedding $\mathbf{v}$:

$$\text{Cosine Similarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \quad (2)$$

This similarity score represents the most semantically similar

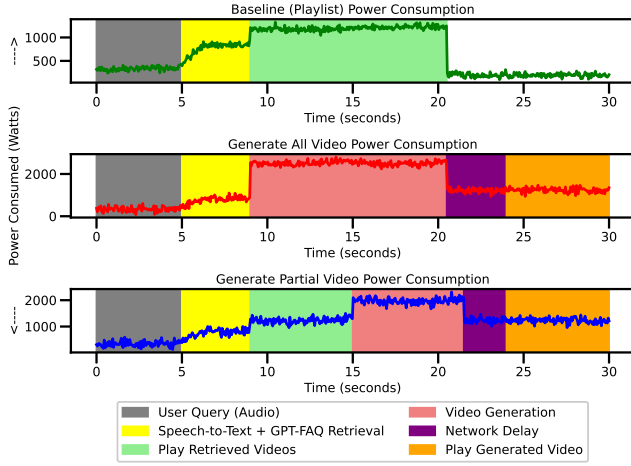


Figure 4: Power consumption across adaptive response modes in *ArtSpeak*, including exact match (pre-generated video), partial match (stitched segments), and no match (real-time video generation).

FAQ, which is then used to fetch and display the corresponding pre-generated video response within the AR interface. This embedding-based retrieval framework ensures contextually accurate low-latency responses at runtime.

4.3 Adaptive Response Modes

The system’s response generation dynamically adapts to the user’s query by evaluating it against a list of precomputed FAQs, following one of three response pathways:

1. **Exact Match (User query to FAQ list):**
When a user query exactly matches an FAQ in the list, the system retrieves the corresponding preprocessed video response almost instantaneously (within two second), as shown in Figure. 4. This rapid retrieval minimizes latency and avoids unnecessary processing, delivering an immediate and fluid user experience.
2. **Partial Match(Generate Partial video):**
For queries that partially match entries in the FAQ list, the system constructs a customized response by stitching together relevant video segments associated with closely related. This process, completed within 5 – 6 seconds, ensures that the user receives a coherent answer tailored to their question, with minimal delay, as shown in Figure. 4.
3. **No Match(Generate All):**
When the system cannot find a match in the FAQ list, it defaults to a generic introductory video to engage the user. Simultaneously, the system begins generating a new, customized video response tailored to the query. Producing a 4 – 6 second custom video in this fallback scenario requires approximately 11.5 seconds. The introductory response plays immediately to maintain user engagement, and as soon as the custom video is ready, it continues playback seamlessly. To further improve coverage and reduce future latency, the system can periodically update its FAQ space as new queries arrive, and use ChatGPT to generate them based on recent user inputs. This approach ensures that the user remains engaged with uninterrupted content while the system prepares a relevant response in the background, as shown in Figure. 4. Performance details of response pathways are discussed in the section 6.

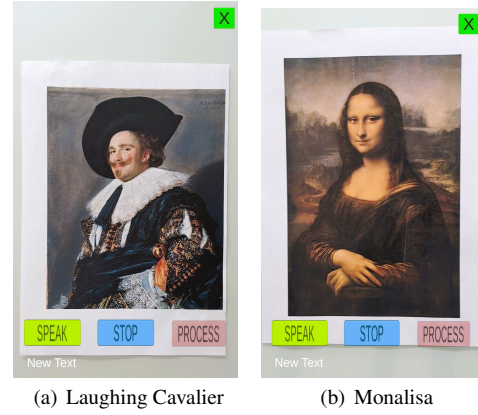


Figure 5: User interface of *ArtSpeak* with auto-triggered interaction buttons upon target image detection

By adapting response pathways based on match type, the system optimizes response times and ensures relevance across a variety of user queries. This structure allows parallel processing to reduce perceived delays, effectively balancing pre-processed and real-time response generation.

5 IMPLEMENTATION

5.1 Generating realistic talking faces

To synthesize realistic talking faces, we used pre-trained GAN models. We use the SADTALKER model [55] to generate expressive talking-head videos from static art portraits. SADTALKER maps input audio to 3D Morphable Model (3DMM) motion coefficients—capturing facial expressions, head pose, and identity features—and generates realistic facial animations using a 3D-aware face renderer. Its ability to produce high-fidelity, expressive outputs without requiring extensive video data makes it well-suited for our application.

We evaluated several alternatives before selecting SADTALKER. Wav2Lip [32] relies on reference videos, limiting its adaptability to various portraits. MakeItTalk [59], which uses 3D landmarks, often results in inaccurate lip-sync for stylized art images. Talking-Face-PC-AVS [57] performs well but heavily dependent on consistent pose videos, making it less flexible for varied artwork. Moreover, Wave2Lip++(SAM-Wav2Lip++) [53] extends basic lip-sync to full body action, by relying on a multistage pipeline-contrastive lip-sync discriminator, SAM-based facial refinement, then a 3D-conv body generator. It has a high inference latency and has limited cross-identity generalization outside of its training domain. EMO’s [44] weakly conditioned latent diffusion produces expressive head motion without 3D proxies, however it needs dozens of denoising steps per frame, leading to longer runtimes and often exhibits temporal flicker without costly post-processing. Diffusion-based lip-synthesis methods such as SyncDiff [13] and Diff2Lip [29] focus diffusion on the mouth region using masked inpainting or bottlenecked temporal priors to enhance synchronization. However, as they rely on multiple sampling steps it proves to be infeasible for real-time use. It may cause boundary seams when merging with the full face. Lastly, the NeRF-based methods, such as AD-NeRF [19] and NeRFFaceSpeech [24], produce photorealistic, view-consistent talking heads; they are unsuitable for interactive deployment due to their heavy per-identity NeRF training or GAN inversion, slow volumetric rendering, and extensive multi-view data. In contrast, SADTALKER supports generalized lip and head movements from a single input image and audio, making it optimal for lifelike

Table 3: User Study Query Evaluation Across Scenarios

Scenario	Total Queries	Fully/Partially Answered	New Query (Generate response)
Instant Match	9	9	0
Multi-Video Playback	9	9	0
Hybrid Real-Time	13	9	4
Total	31	27	4

portrait animation in our AR system.

5.2 AR Framework Selection

To develop the *ArtSpeak* application, we evaluated two augmented reality (AR) frameworks: Unity’s ARFoundation and the Vuforia Engine. The ARFoundation supports cross-platform development for Android and iOS, however, we observed drawbacks in its image tracking performance, mainly with low-feature-point images or artworks.

In contrast, Vuforia offers robust image recognition functionality with built-in support for feature point image detection. The Vuforia Target Manager rates the input images based on feature richness, ensuring reliable tracking and alignment in real-world AR scenarios. We selected Vuforia for its consistent tracking performance across art portraits with different texture.

5.3 AR Application Development

We uploaded six selected portrait images of the Mona Lisa, Girl with a Pearl Earring, Adele (Portrait of Adele Bloch-Bauer I), The Blue Boy, Frida (Self-Portrait with Thorn Necklace and Hummingbird), and The Laughing Cavalier to the Vuforia Target Manager. Each image was ranked for detectability, and high-scoring targets were linked to corresponding pre-generated talking-face videos. In particular, the Laughing Cavalier showed seamless detection due to its high feature-point density, whereas portraits with fewer feature points maintained stable overlays with minimal distortion during video playback.

Within Unity 2021.3 and the Vuforia AR SDK framework, we configured AR Camera and Image Target objects in the scene. The talking videos generated using SADTALKER were mapped to these targets using placeholder 3D quads for spatial alignment. The final application was deployed and tested on Android devices, showing robust performance in varying lighting conditions, camera angles, and distances.

5.4 Improvement usability

To provide seamless interaction, we incorporated "Speak," "Process," and "Stop" buttons into the application as shown in Figure. 5. The 'Speak' button allows users to ask questions with audio input, which is converted to text. The application retrieves and plays a carefully matched response for the user, on clicking the 'Process' button. The 'Stop' button allows users to terminate the speaking art portrait once their query is answered. Furthermore, users can also stop the interaction at any time to explore other artworks in the museum.

6 EXPERIMENTAL ANALYSIS

We conducted a user study with 23 participants (15 male and 8 female), aged between 18 and 35, to evaluate the usability, responsiveness, and interaction quality of *ArtSpeak* application. Participants were recruited to ensure a diverse range of familiarity with augmented reality (AR), reported on a 5-point scale (1 = Not familiar, 5 = Very familiar). Majority of the participants reported medium to high technological proficiency and had a varying frequency of museum visits, like weekly, monthly, and yearly(rare)

visits. This demographic range allowed us to assess the usability of the application between frequent and infrequent museum visitors, as well as across different levels of AR experience.

6.1 User Study Design and Tasks

Participants were introduced to *ArtSpeak* mobile AR application and received a brief introduction to its interface and functionalities. The study was organized around a list of scenario-based tasks that were similar to realistic museum interactions using an art *The Laughing Cavalier* by Frans Hals.

The task was designed to evaluate the system’s ability to process a range of user queries, divided into three scenarios:

1. Instant Match:

This scenario assessed the application’s ability to respond with pre-processed video that immediately matched the user’s query and displayed on the application screen.

2. Multi-Video Playback:

It checks the application’s ability to play two or more pre-computed videos sequentially to answer a compound query and display them on the application screen.

3. Hybrid Real-Time:

This case evaluates the application’s ability to play a pre-computed video as an introduction while simultaneously generating a tailored response in real time, which is then displayed on the application screen. Additionally, when the user query contains both predefined and dynamic components, the system instantly retrieves responses for the predefined parts while generating content for the dynamic new segments. This hybrid approach is designed to minimize latency and enhance user engagement.

The graph plots in Figure. 6 display the users’ satisfaction levels across the evaluation metrics discussed in Section 6.2 for each of the user query scenarios and their respective responses. The first plot for *Instant Match* indicates high user satisfaction across all evaluation metrics, aligning with our expectations when user queries directly match pre-generated video responses. The second plot for *Multi-Video Playback* demonstrates similarly strong satisfaction levels, even when the system retrieves and plays multiple videos sequentially to form a complete response, although few users reported a slight increase in perceived lag. Surprisingly, in the *Hybrid Real-Time* condition, which we expected to be more challenging, most users reported being either “satisfied” or “neutral” across key metrics such as reduced lag, smooth video playback, and overall experience. These results confirm the system’s ability to maintain effectiveness and user engagement even when partially generating responses in real time, establishing *ArtSpeak* as a practical and efficient application for art museum experiences.

6.2 User study evaluation

After interacting with the application for each of the scenarios, the participants answered a feedback survey to rate their experience. The survey had the following evaluation metrics per scenario:

- Query Satisfaction:** How well did the response(s) address your query? Options: *Very Satisfied, Satisfied, Neutral, Dissatisfied, Very Dissatisfied*
- Lag Observation:** Did you observe any delay or lag in response playback? Options: *No lag, Minimal lag, Noticeable lag, Significant lag*
- Playback Smoothness:** For scenarios with multiple video responses, did the videos play sequentially and smoothly? Options: *Very Smooth, Smooth, Neutral, Slightly Disjointed, Disjointed*

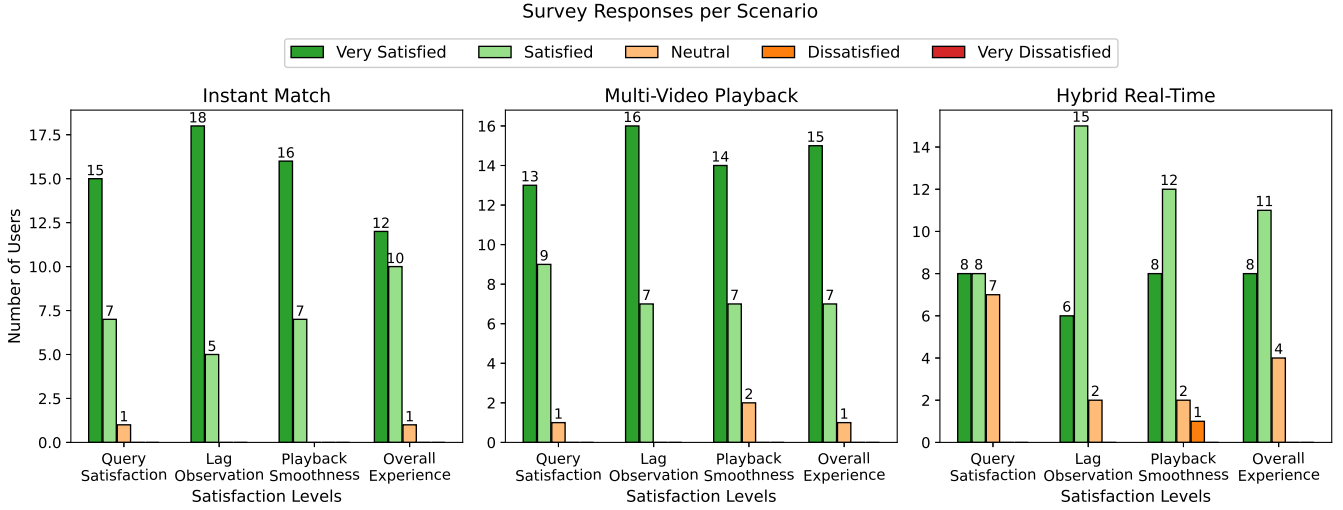


Figure 6: User satisfaction across three query scenarios, evaluated by Query Satisfaction, Lag Observation, Playback Smoothness, and Overall Experience.

Table 4: Evaluation Metrics from User Study of the *ArtSpeak* Application

Metric	Description	Result
Task Satisfaction (1–5)	Average satisfaction with answers provided in each scenario.	4.3
System Responsiveness (Lag) (1–5)	Perceived lag or delay when viewing content (higher rating = minimal lag).	4.03
Video Smoothness (1–5)	How smoothly the scenario videos played.	4.42
Overall Experience (1–5)	Overall experience of each scenario.	4.4
Sample Size	Number of participants in the study.	23

Table 5: Mean user ratings with standard deviation and bootstrap-based confidence intervals

Metric	Scenario	Mean	Std Dev	Confidence Interval
Query Satisfaction	Instant	4.61	0.58	0.48
	Multi	4.48	0.73	0.61
	Hybrid	3.96	0.98	0.78
Lag Observation	Instant	3.13	1.01	0.78
	Multi	3.43	0.79	0.65
	Hybrid	3.04	0.87	0.65
Smooth Playback	Instant	4.61	0.61	0.57
	Multi	4.48	0.72	0.65
	Hybrid	4.17	0.69	0.61
Overall Experience	Instant	3.48	0.67	0.48
	Multi	3.61	0.72	0.43
	Hybrid	3.17	0.87	0.61

4. Overall Experience: Please rate your overall experience for each scenario. Options: *Poor, Average, Good, Excellent*

The survey responses show high user satisfaction for all three scenarios, with 85% of participants finding responses relevant to the queries. Only 10–15% of them observed noticeable lag, indicating efficient retrieval of responses. In multi-video playback scenarios, 90% of participants rated the video playback smoothness as good or excellent. The Table 4 describes the evaluation of our user study, thus highlights the system’s ability to provide an intuitive and engaging AR interface. The user-study observations (N=23) were statistically evaluated with confidence intervals for every mean rating as shown in Table. 5. The *confidence intervals* shows that the user ratings do not vary significantly; most users rate the query satisfaction, lag, and smoothness highly, especially for the instant-match and multi-video settings.

6.3 System Accuracy and Latency Evaluation

We compared the retrieval-based *ArtSpeak* pipeline against a baseline that performs real-time video generation using the SADTALKER model, to evaluate the system performance. As shown in Table. 6, full video generation took 60–90 seconds on a GPU-equipped desktop, and up to 6 minutes on a CPU-only machine. In contrast, *ArtSpeak* achieved faster response times, with an average latency of approximately 2.5 seconds in the direct match

scenario, 5.5 seconds for playlist-based retrieval, and 11.5 seconds in the partial generation case.

Therefore, compared to the baseline, our application achieved a latency improvement of more than 30× in the best-case scenario (~2–3 seconds retrieval), and at least 5× in the worst-case scenario (~10–12 seconds retrieval), when compared to real-time video generation using SADTALKER model on a GPU. Additionally, our system demonstrated significantly lower power consumption during inference.

6.4 Evaluation of FAQ Retrieval Strategies

We implemented and evaluated five different retrieval approaches by varying embedding models, similarity metrics, and indexing techniques to evaluate their effectiveness. For each configuration, we maintained query relevance judgment files (Qrels) to evaluate how accurately and consistently the retrieved responses mapped user queries to the expected FAQ answers. These Qrels files were used to compute standard information retrieval performance metrics, for comparative analysis of retrieval quality.

Each retrieval system was evaluated using the following metrics:

- **Precision:** The proportion of relevant FAQs retrieved among the top-ranked results.

Table 6: Latency Comparison Between *ArtSpeak* Scenarios and SadTalker Baseline

Scenario	Latency (sec)
Scenario 1: Direct Match (Instant video response)	~2–3 sec
Scenario 2: Playlist Match (multi-video response)	~5–6 sec
Scenario 3: Generate Partial Video (video response + partial video generation)	~11–12 sec
Baseline (Real-Time video generation using SadTalker)	60–90 sec (GPU) / 180–360 sec (CPU)

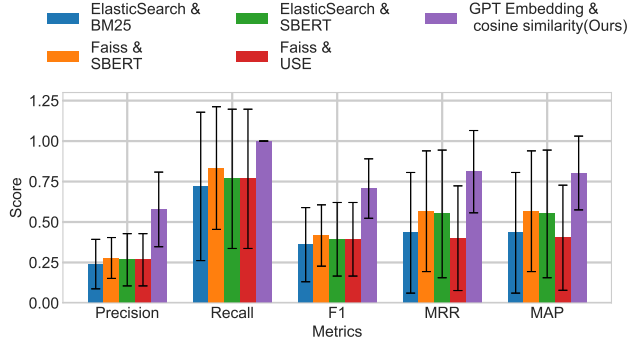


Figure 7: Performance Metrics Assessment for FAQ Retrieval Models"

- **Recall:** The proportion of relevant FAQs retrieved out of all relevant ones available.
- **F1 Score:** The harmonic mean of precision and recall, reflecting the balance between the two.
- **Mean Reciprocal Rank (MRR):** The inverse rank of the first correct response, averaged over all queries.
- **Mean Average Precision (MAP):** The average of precision scores across all relevant FAQ retrievals for each query.

In our system, we explore two ways for matching user queries with FAQ responses. The first approach involves scoring all FAQs against a given query using semantic similarity. We use several language models, such as SBERT, USE, and OpenAI’s GPT embeddings, to convert sentences into vector representations. We then calculate the cosine similarity between the query and each FAQ to score similarity measure based on how closely they match. Additionally, we also used BM25 [35] word based similarity score to check and compare its performance with embedding based approaches. Since our FAQ collection is relatively small, we store these embeddings in lightweight files and efficiently compare all FAQs in real time. Among all models, GPT-3 embeddings combined with cosine similarity achieved the best performance in terms of accuracy and contextual understanding, without requiring a dedicated indexing system.

We also use FAISS [11], a fast vector search library, to retrieve the most relevant FAQs without scoring all entries directly. FAISS returns the top-k closest matching FAQs from the embedding space. We tested FAISS with both SBERT and USE embeddings. While FAISS offers faster retrieval and better scalability, we found that the full scoring method using GPT embeddings performed better for our domain-specific queries.

Table 7: Paired significance of GPT +CosSim vs. other retrieval methods. Bold Δ indicates GPT performs better; $\dagger$ marks $p < .05$.

Compared methods	Metric	Δ	p_t	p_w
BM25	Precision	+0.360[†]	<0.0001	<0.0001
	Recall	+0.280 [†]	0.0054	0.0082
	F1	+0.364[†]	<0.0001	<0.0001
	MRR	+0.273 [†]	0.0115	0.0208
Faiss + USE	Precision	+0.311[†]	<0.0001	<0.0001
	Recall	+0.233 [†]	0.0059	0.0082
	F1	+0.313[†]	<0.0001	<0.0001
	MRR	+0.250 [†]	0.0036	0.0059
ES + SBERT	Precision	+0.311[†]	<0.0001	<0.0001
	Recall	+0.233 [†]	0.0059	0.0082
	F1	+0.313[†]	<0.0001	<0.0001
	MRR	+0.328[†]	0.0002	0.0006
Faiss + SBERT	Precision	+0.300[†]	<0.0001	<0.0001
	Recall	+0.167 [†]	0.0226	0.0253
	F1	+0.290[†]	<0.0001	<0.0001
	MRR	+0.200 [†]	0.0301	0.0476

To evaluate these retrieval methods, we used a set of 30 user queries and retrieved the top-matching FAQs as a response from a pool of 140 FAQs. We then compared each query with the top FAQs retrieved by the system and labeled whether each FAQ was relevant or not. The evaluation results across the five metrics with their mean scores, standard deviations, and confidence intervals(CI) are shown in Table.8. The traditional BM25 approach achieves reasonable coverage (recall=0.72), however, it results in a high false positive rate (precision = 0.24). Moreover, when sentence embeddings are used such as Faiss+SBERT, ES+SBERT, or Faiss+USE, the precision value rises up to about 0.28, and recall rises slightly (0.83). This confirms the precision-vs-recall trade-off in dense retrieval. In comparison, the GPT-based embeddings overcome the above-mentioned limitation. It finds all relevant FAQs (recall =1) and has almost double the precision achieved using BM25 (0.58).This leads to an F1 score of 0.71 and a better ranking performance (MRR = 0.81, MAP = 0.80), which means the users get correct responses quickly and more reliably. Based on the evaluation, *GPT-based embedding with cosine similarity and dictionary indexing(Ours)* consistently outperformed all other methods across all metrics. To further confirm the statistical significance of *GPT-based embedding with cosine similarity and dictionary indexing(Ours)*, the retrieval methods were evaluated using paired t-tests and Wilcoxon signed-rank tests over shared queries(Table 7). The tests show that the improvements due to *GPT-based embedding (Ours)* are statistically significant.

The results summarized in Figure. 7 verified the ideal balance between retrieval accuracy and inference latency using large language model embeddings (GPT) with similarity-based dictionary lookup. Hence, this approach was integrated into our system.

6.5 Power Consumption Analysis: Mobile Application vs. Execution of Deep Learning Models

We evaluated the energy efficiency of the *ArtSpeak* application by comparing two methods of deployment: real-time execution of the SADTALKER model versus offline integration of pre-generated video responses. In the real-time talking-face generation scenario, the SADTALKER model required over one minute to generate each talking-head video, resulting in noticeable latency and high energy consumption. Meanwhile, the offline approach allowed immediate playback by retrieving pre-generated videos and displaying them onto the target artwork in the Unity-based AR interface. To compare the power usage, we used Battery Historian on Android to monitor the mobile application, which consumed approximately 5.4

Table 8: Mean, standard deviation, confidence interval for each retrieval system

Metric	System	Mean	Std Dev	Confidence Interval
Precision	GPT + CosSim	0.58	0.23	0.08
	Faiss + USE	0.27	0.16	0.06
	ES + SBERT	0.27	0.16	0.06
	Faiss + SBERT	0.28	0.13	0.05
	ES + BM25	0.24	0.15	0.06
Recall	GPT + CosSim	1.00	0.00	0.00
	Faiss + USE	0.77	0.43	0.15
	ES + SBERT	0.77	0.43	0.15
	Faiss + SBERT	0.83	0.38	0.14
	ES + BM25	0.72	0.46	0.18
F1 Score	GPT + CosSim	0.71	0.18	0.07
	Faiss + USE	0.39	0.23	0.08
	ES + SBERT	0.39	0.23	0.08
	Faiss + SBERT	0.42	0.19	0.07
	ES + BM25	0.36	0.23	0.09
MRR	GPT + CosSim	0.81	0.25	0.09
	Faiss + USE	0.40	0.32	0.12
	ES + SBERT	0.55	0.39	0.14
	Faiss + SBERT	0.57	0.37	0.13
	ES + BM25	0.43	0.37	0.15
MAP	GPT + CosSim	0.80	0.23	0.08
	Faiss + USE	0.40	0.32	0.12
	ES + SBERT	0.55	0.39	0.14
	Faiss + SBERT	0.57	0.37	0.13
	ES + BM25	0.43	0.37	0.15

Note: ES = ElasticSearch, SBERT = Sentence-BERT, USE = Universal Sentence Encoder, CosSim = cosine similarity.

watts and 593.70 joules during one-minute usage sessions. Furthermore, Intel Power Gadget 3.6 was used on Windows to monitor the execution of the SADTALKER model, showing high energy consumption of 3099.49 joules for generating a single video. The retrieval-based mobile application achieved an approximately 81% reduction in energy consumption compared to real-time video generation using SADTALKER on a desktop. These observations confirm that the offline pre-generation strategy significantly improves responsiveness and reduces energy demands, making the system suitable for deployment on mobile devices.

7 LIMITATIONS

We designed *ArtSpeak* to provide an immersive museum experience for users, while being scalable and energy efficient. However, the current iteration of the work does have the following limitations, that we will address as part of future work. First, although our extensive evaluation and user study demonstrated strong user engagement, it was conducted using a single artwork. This limits the generalizability in more diverse museum settings. Second, the FAQ database will continue to grow as more user interactions occur and additional artworks are incorporated into the system. To ensure efficient long-term maintenance, frequently asked queries should be retained for immediate access, while rarely accessed ones could be removed. In certain edge cases, such as handling ambiguous user queries that lack clear intent, the system will find it difficult to retrieve the answer accurately and may require additional prompting from the user. Currently, the system supports interactions only in English; however, it could be extended to support region-specific languages to better engage users from different countries and cultural backgrounds.

8 FUTURE WORKS

First, improve real-time video creation to reduce latency and elevate the user experience. Furthermore, incorporate vast language

assistance with culturally accurate lip-syncing and customization through natural language understanding (NLU). This could facilitate customized interactions, enhance accessibility, and promote inclusivity.

In addition to the system’s effectiveness in museums it can be extended to various other domains. For instance, it could be used in poster presentations by embedding video explanations of research work. Individuals could scan a presenter’s picture on a poster to view their recorded presentation. Similarly, interactive business cards could incorporate videos that offer detailed introductions to an individual’s work. Finally, the application could enhance AR video calls, enabling lifelike interactions that simulate physical presence.

Furthermore, teachers can also use our system to create interactive lessons, bringing historical figures or language tutors to life without specialized hardware. Additionally, learners with hearing impairments can lip-read the talking-head video content, which can be paired with real-time captioning. Learners with vision impairment can interact with the application with audio queries and audio responses from the video content.

In the next stage, our current system can be improved by exploring unsupervised personalization, which enables adaptive user experiences without the need for annotated data. Moreover, integrating real-time generative dialogue models may improve the interactivity of the system by allowing contextually aware dynamic responses aligned with the intent of the user.

In the future, we can address these improvements and explore different use cases like virtual tourism, remote education, and cultural storytelling. This could lead to a robust and scalable application for interactive learning, communication, and engagement across both professional and educational settings.

9 CONCLUSION

We present a novel augmented reality application for museums and art galleries that brings portraits to life in response to user queries. At its core *ArtSpeak* interprets user query and retrieves closely aligned information about the artwork in the form of talking head video responses to facilitate an interactive experience for visitors. For smooth tracking and rendering of animated information, it integrates generative AI and natural language processing (NLP) methods in Unity framework. Its user-friendly design and minimal computational requirements make it easily accessible on commodity smartphones. Our application turns the conventional museum into an engaging experience with its curated responses, dependable image tracking, intuitive scanning, and real-time conversational engagement. This work demonstrates how AR technology integrated with an efficient information retrieval pipeline can improve visitors’ interactions with artworks in museum environments.

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